

PHYS 580 Homework 4 (Chap.5)

All questions are worth 10 points, irrespective of complexity or length. Please submit all your solutions on Brightspace in PDF format, together with the source code (if any). Make sure to

- a) discuss the physics and your results, as appropriate and/or directed,
- b) include graphical output to illustrate the results,
- c) include the source codes of your programs (at least the critical parts thereof),
- d) describe briefly what your program does, and how it does it (algorithm), and
- e) state the nature of the numerical approximation you used and demonstrate how you know that the approximation (with the parameters you used) is adequate for the problem you used it for.

Note the errata pages for the textbook: <http://www.physics.purdue.edu/~hisao/book/www/errata.pdf>
which are explained further on <http://www.physics.purdue.edu/~hisao/book/www/errata.html>

1. Problem 5.7 (p.143).

Write two programs to solve the capacitor problem of Figures 5.6 and 5.7, one with the Jacobi method and one using the SOR algorithm. For a fixed accuracy (set by the convergence test) compare the number of iterations, N_{iter} , that each algorithm requires as a function of the number of grid elements, L . Show that for the Jacobi method $N_{\text{iter}} \sim L^2$, while with SOR $N_{\text{iter}} \sim L$.

Note that you can borrow most of the work from Lab 7, but now you must also include the SOR algorithm in your comparisons.

2. Problem 5.8 (p.147).

Extend our treatment of a point charge in a metal box to deal with the case in which the charge is located near one face of the box. Study how the equipotential contours are affected by the proximity of a grounded surface (the face of the box).

3. Problem 5.11 (p.151).

Calculate the magnetic field from a straight wire of current using Simpson's rule (see Appendix E), and compare it with the result obtained from (5.25) for the same grid size, Δz .